

# ALMOST SUBSPACES OF $\mathbb{F}_2^n$ : A CORRECTED AND COMPLETE PROOF

ABSTRACT. Let  $\mathbb{F}_2^n$  be equipped with the normalized Hamming metric  $d(x, y) = \frac{1}{n} |\{i \mid x_i \neq y_i\}|$ . We prove that for every  $\varepsilon > 0$  there exists  $\delta > 0$ , independent of  $n$ , such that every non-empty  $C \subset \mathbb{F}_2^n$  satisfying  $C + C \subset C_\delta$  lies within Hausdorff distance  $\varepsilon$  of some affine subspace of  $\mathbb{F}_2^n$ .

This document also gives a careful analysis of a gap in a previous ultraproduct-based argument, and presents two complete proofs: one via a corrected ultraproduct argument that properly transfers the subspace structure from the limit back to the factors, and one self-contained combinatorial proof via greedy bases that avoids all non-elementary tools.

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## 1. THE MAIN THEOREM AND OVERVIEW

**Definition 1.1.** For  $C \subset \mathbb{F}_2^n$  and  $\alpha \geq 0$ , write  $C_\alpha = \{y \in \mathbb{F}_2^n : \exists x \in C, d(x, y) \leq \alpha\}$ . The *Hausdorff distance* is  $D(X, Y) = \inf\{\alpha \geq 0 : X \subset Y_\alpha \text{ and } Y \subset X_\alpha\}$ .

**Theorem 1.2** (Main Theorem). *For every  $\varepsilon > 0$  there exists  $\delta = \delta(\varepsilon) > 0$  such that for all  $n \in \mathbb{N}$ , every non-empty  $C \subset \mathbb{F}_2^n$  satisfying  $C + C \subset C_\delta$  satisfies  $D(C, \mathcal{A}) \leq \varepsilon$  for some affine subspace  $\mathcal{A} \leq \mathbb{F}_2^n$ .*

We give two complete proofs. The first (Section 3) uses ultraproducts and requires care about how subspace structure transfers from the ultraproduct back to the factors. The second (Section 5) is a self-contained combinatorial proof.

## 2. THE GAP IN THE NAIVE ULTRAPRODUCT ARGUMENT

A natural approach to Theorem 1.2 runs as follows.

**Naive argument.** Suppose the theorem fails for some  $\varepsilon_0 > 0$ . Then for each  $k$  there exist  $n_k$  and  $C_k \subset \mathbb{F}_2^{n_k}$  with  $C_k + C_k \subset (C_k)_{1/k}$  but  $D(C_k, \mathcal{A}) > \varepsilon_0$  for every affine subspace  $\mathcal{A} \leq \mathbb{F}_2^{n_k}$ .

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Fix a non-principal ultrafilter  $\omega$  on  $\mathbb{N}$  and form the ultraproduct  $C_\omega = \prod_{k \rightarrow \omega} C_k$  with metric  $d_\omega([(x_k)], [(y_k)]) = \lim_{k \rightarrow \omega} d_k(x_k, y_k)$ .

Since  $C_k + C_k \subset (C_k)_{1/k}$ , for any  $[(x_k)], [(y_k)] \in C_\omega$  there exist  $z_k \in C_k$  with  $d_k(x_k + y_k, z_k) \leq 1/k$ , and  $\lim_{k \rightarrow \omega} d_k(x_k + y_k, z_k) = 0$ . So  $[(x_k)] + [(y_k)] = [(z_k)] \in C_\omega$ .

Assuming  $0 \in C_k$  (by translating),  $c + c = 0 \in C_k$  for all  $c \in C_k$  (over  $\mathbb{F}_2$ ), so  $C_\omega$  is an  $\mathbb{F}_2$ -vector space.

**Claim (naive):** Since  $C_\omega$  is a subspace of  $(\mathbb{F}_2^{n_k})_\omega$ , for  $\omega$ -almost every  $k$  there are subspaces  $V_k \leq \mathbb{F}_2^{n_k}$  with  $D(C_k, V_k) \rightarrow 0$ .

**Remark 2.1** (The gap). This claim is *not justified as stated*, and this is the critical gap in the naive argument. The issue is two-fold.

(i)  **$C_\omega$  can be infinite-dimensional.** The ultraproduct  $(\mathbb{F}_2^{n_k})_\omega$  is a huge non-standard object;  $C_\omega$  is an  $\mathbb{F}_2$ -subspace of it and can be infinite-dimensional. For example, take  $C_k = \mathbb{F}_2^{n_k}$  (the whole space): then  $C_\omega = (\mathbb{F}_2^{n_k})_\omega$  is infinite-dimensional. A subspace of an infinite-dimensional ultraproduct need not descend to subspaces of the factors.

(ii) **Even if  $C_\omega$  were finite-dimensional, the transfer needs work.** If  $C_\omega = \text{span}_{\mathbb{F}_2}([(c_1^{(k)})], \dots, [(c_r^{(k)})])$  for some elements with  $c_j^{(k)} \in C_k$ , one might set  $V_k = \text{span}(c_1^{(k)}, \dots, c_r^{(k)})$ . But we must verify that:

- (a)  $C_k \subset (V_k)_\varepsilon$  for  $\omega$ -a.e.  $k$ , and
- (b)  $V_k \subset (C_k)_\varepsilon$  for  $\omega$ -a.e.  $k$ .

Part (a) requires that every element of  $C_k$  is  $\varepsilon$ -close to  $V_k$ , meaning every equivalence class in  $C_\omega$  is  $\varepsilon$ -close to  $C_\omega$  itself — which is trivially true *for*  $C_\omega$  but requires a non-trivial finite approximation argument to descend to the  $C_k$ . Part (b) requires every element of  $V_k$  to be  $\varepsilon$ -close to  $C_k$ . An element of  $V_k$  is a sum of at most  $r$  elements of  $\{c_1^{(k)}, \dots, c_r^{(k)}\} \subset C_k$ , so by the iterated addition lemma it lies in  $(C_k)_{(r-1)/k}$ . This tends to 0 along  $\omega$  only if  $r$  is fixed (independent of  $k$ ). If  $r$  were allowed to grow with  $k$ , this argument would fail.

**Summary of what is needed.** To make the ultraproduct argument work, one must show that  $C_\omega$  is *finite-dimensional* as an  $\mathbb{F}_2$ -vector space, with dimension depending only on  $\varepsilon$  (not on  $k$  or  $n_k$ ). This is equivalent to showing that  $C_k$  has a net of size bounded uniformly in  $k$  — which is the content of Lemma 3.1 below.

### 3. COMPLETE PROOF VIA ULTRAPRODUCTS (CORRECTED)

We now give the corrected ultraproduct proof. The key additional ingredient is a uniform covering number bound.

#### 3.1. A uniform metric-dimension bound.

**Lemma 3.1** (Uniform covering bound). *Let  $C \subset \mathbb{F}_2^n$  be non-empty with  $0 \in C$ , and suppose  $C + C \subset C_\delta$ . Then for every  $\alpha > 0$  with  $\alpha > 2\delta$ :*

$$N(C, \alpha) \leq N(C, \frac{\alpha - \delta}{2})^2,$$

where  $N(C, r)$  denotes the minimum number of  $r$ -balls needed to cover  $C$ .

In particular, setting  $\alpha = 3\delta$  gives  $N(C, 3\delta) \leq N(C, \delta)^2$ , so the covering numbers at scale  $3\delta$  are controlled by those at scale  $\delta$ .

*Proof.* Let  $M = N(C, \frac{\alpha-\delta}{2})$  and let  $x_1, \dots, x_M \in C$  be an  $\frac{\alpha-\delta}{2}$ -net for  $C$ , so  $C \subset \bigcup_i B(x_i, \frac{\alpha-\delta}{2})$ .

We claim the  $M^2$  elements  $\{x_i + x_j : 1 \leq i, j \leq M\}$  form an  $\alpha$ -net for  $C + C$  (and hence for  $C + C \subset C_\delta$ ).

For any  $c_1, c_2 \in C$ , pick  $i, j$  with  $d(c_1, x_i) \leq \frac{\alpha-\delta}{2}$  and  $d(c_2, x_j) \leq \frac{\alpha-\delta}{2}$ . Then  $d(c_1 + c_2, x_i + x_j) = d(c_1 + c_2 + x_i + x_j, 0) \leq d(c_1 + c_2, x_i + x_j) \leq d(c_1, x_i) + d(c_2, x_j) \leq \alpha - \delta$ .

Since  $c_1 + c_2 \in C + C \subset C_\delta$ , there exists  $c^* \in C$  with  $d(c_1 + c_2, c^*) \leq \delta$ . By the triangle inequality,  $d(c^*, x_i + x_j) \leq (\alpha - \delta) + \delta = \alpha$ . So every element of  $C \supset C_\delta \supset C + C$  is within distance  $\alpha$  of some  $x_i + x_j$ .

Thus  $N(C, \alpha) \leq M^2 = N(C, \frac{\alpha-\delta}{2})^2$ .  $\square$

**Remark 3.2.** Lemma 3.1 does *not* by itself give a bound on  $N(C, \alpha)$  independent of  $n$ . It gives a self-referential bound: the covering number at one scale controls the covering number at a smaller scale. What we actually need is a separate argument bounding  $N(C, \delta)$  (the covering number at the *same* scale as the hypothesis). This is where the structure of  $\mathbb{F}_2^n$  and the normalized metric play a role.

### 3.2. Why $C_\omega$ is finite-dimensional.

**Lemma 3.3.** *In the ultraproduct setting of Section 2 (with  $0 \in C_k$  and  $C_k + C_k \subset (C_k)_{1/k}$ ), the vector space  $C_\omega$  is finite-dimensional over  $\mathbb{F}_2$ .*

*More precisely, for each fixed  $\alpha > 0$ , the covering number  $N(C_\omega, \alpha)$  is finite.*

*Proof.* Suppose  $N(C_\omega, \alpha) \geq m$  for some  $m$ . Then there exist  $m$  elements  $[(p_1^{(k)})], \dots, [(p_m^{(k)})] \in C_\omega$  with  $d_\omega([(p_i^{(k)})], [(p_j^{(k)})]) > \alpha$  for  $i \neq j$ , i.e.,  $\lim_{k \rightarrow \omega} d_k(p_i^{(k)}, p_j^{(k)}) > \alpha$ .

In particular, for  $\omega$ -almost all  $k$ ,  $d_k(p_i^{(k)}, p_j^{(k)}) > \alpha/2$  for all  $i \neq j$ . So  $\{p_1^{(k)}, \dots, p_m^{(k)}\}$  is an  $(\alpha/2)$ -separated set in  $C_k \subset \mathbb{F}_2^{n_k}$ .

The Hamming metric space  $\mathbb{F}_2^{n_k}$  can accommodate arbitrarily large separated sets (take  $m = n_k$ ), so this does *not* immediately bound  $m$ .

However, we can bound  $m$  using the hypothesis  $C_k + C_k \subset (C_k)_{1/k}$  together with the following key observation.  $\square$

**Remark 3.4** (The correct approach to finite-dimensionality). The argument for finite-dimensionality of  $C_\omega$  cannot be purely metric-ultraproduct: it must use the additive hypothesis more substantially.

The following is the correct route. We will prove directly (without ultraproducts) the bound  $N(C, \alpha) \leq \exp(C_0/\alpha^2)$ , uniformly in  $n$ , for any  $C$  with  $C + C \subset C_\delta$  with  $\delta$  small enough. This is the content of Lemma 3.5 below. Given this,  $N(C_k, \alpha)$  is bounded uniformly in  $k$  for each fixed  $\alpha$ , so the ultraproduct  $C_\omega$  has finite covering number at every scale, hence is finite-dimensional.

**Lemma 3.5** (Uniform net bound via additive energy). *Let  $C \subset \mathbb{F}_2^n$  with  $|C| = N$  and  $C + C \subset C_\delta$ . Then for any  $\alpha > 4\delta$ :*

$$N(C, \alpha) \leq \frac{4}{\alpha^2}.$$

*Proof.* Let  $S = \{s_1, \dots, s_m\} \subset C$  be a maximal  $\alpha$ -separated set. Every  $s_i \in C$ , and for  $i \neq j$ ,  $d(s_i, s_j) > \alpha$ .

We embed  $C$  (hence  $S$ ) in  $\mathbb{R}^n$  by identifying  $\mathbb{F}_2^n \equiv \{0, 1\}^n$  and normalizing: let  $f_i = s_i/n^{1/2} \in \mathbb{R}^n$  (viewing  $s_i \in \{0, 1\}^n$ ). Actually it is cleaner to work with the  $\pm 1$ -embedding: define  $\phi : \mathbb{F}_2^n \rightarrow \{-1, +1\}^n$  by  $\phi(x)_i = 1 - 2x_i$ .

Then for  $x, y \in \mathbb{F}_2^n$ :

$$\frac{1}{n}\phi(x) \cdot \phi(y) = \frac{1}{n} \sum_{i=1}^n (1 - 2x_i)(1 - 2y_i) = 1 - 2d(x, y).$$

(Here  $\cdot$  is the standard inner product on  $\mathbb{R}^n$ .) So  $d(x, y) > \alpha \iff \frac{1}{n}\phi(x) \cdot \phi(y) < 1 - 2\alpha$ .

Set  $g_i = \frac{1}{\sqrt{n}}\phi(s_i) \in \mathbb{R}^n$ . Then  $\|g_i\|^2 = 1$  and  $\langle g_i, g_j \rangle = \frac{1}{n}\phi(s_i) \cdot \phi(s_j) = 1 - 2d(s_i, s_j)$ . Since  $d(s_i, s_j) > \alpha$  for  $i \neq j$ :

$$\langle g_i, g_j \rangle < 1 - 2\alpha, \quad i \neq j.$$

Consider  $F = g_1 + g_2 + \dots + g_m \in \mathbb{R}^n$ . Then

$$0 \leq \|F\|^2 = \sum_i \|g_i\|^2 + \sum_{i \neq j} \langle g_i, g_j \rangle < m + m(m-1)(1-2\alpha) = m[1 + (m-1)(1-2\alpha)].$$

This gives  $1 + (m-1)(1-2\alpha) > 0$ , i.e.,  $(m-1)(1-2\alpha) > -1$ , which for  $\alpha < 1/2$  gives  $m-1 < \frac{1}{2\alpha-1} \dots$  this is the wrong direction when  $\alpha < 1/2$ .

Let us redo this. For  $\alpha < 1/2$ ,  $1 - 2\alpha > 0$ , so the cross terms are positive and  $\|F\|^2 \geq 0$  gives no upper bound on  $m$ .

We need a better approach for the regime  $\alpha \in (0, 1/2)$ .

**Corrected approach.** Use the  $\mathbb{F}_2$ -structure. The  $\pm 1$  embedding is useful for  $\alpha > 1/2$  but not for the regime we care about.

Instead, we use the following direct argument.

For  $s_i, s_j \in S \subset C$ , the sum  $s_i + s_j \in C + C \subset C_\delta$ , so there exists  $c_{ij} \in C$  with  $d(s_i + s_j, c_{ij}) \leq \delta$ .

Now the elements  $c_{ij}$  (for  $i < j$ ) are elements of  $C$ . Two distinct pairs  $(i, j) \neq (i', j')$  can give the same  $c_{ij} = c_{i'j'}$  only if  $s_i + s_j$  and  $s_{i'} + s_{j'}$  are both close to the same element of  $C$ . Specifically,  $d(s_i + s_j, s_{i'} + s_{j'}) \leq 2\delta$ .

But by the separation of  $S$ :

$$\begin{aligned} d(s_i + s_j, s_{i'} + s_{j'}) &= d(s_i + s_j + s_{i'} + s_{j'}, 0) \\ &\geq (\text{some lower bound in terms of } \alpha) \end{aligned}$$

... but this lower bound is not straightforward.

The key insight is that the elements  $\{s_i + s_j : i < j\} \cup \{s_i : i\}$  form an almost-group and we can use energy counting.

**Correct bound via Gram matrix.** Return to the Gram matrix approach, but use it correctly.

Let  $G$  be the  $m \times m$  Gram matrix with  $G_{ij} = \langle g_i, g_j \rangle = 1 - 2d(s_i, s_j)$ .  $G$  is positive semidefinite. In particular  $\det(G) \geq 0$  and all eigenvalues of  $G$  are non-negative.

The trace of  $G$  equals  $m$  (since  $G_{ii} = 1$ ). All off-diagonal entries satisfy  $G_{ij} = 1 - 2d(s_i, s_j) \leq 1 - 2\alpha$  for  $i \neq j$  (since  $d(s_i, s_j) > \alpha$ ).

Also,  $G_{ij} \geq -1$  always (since  $d \leq 1$ ).

The minimum eigenvalue of  $G$  satisfies (by Gershgorin's circle theorem):

$$\lambda_{\min}(G) \geq G_{ii} - \sum_{j \neq i} |G_{ij}| \geq 1 - (m-1).$$

This gives a lower bound on eigenvalues but not a useful upper bound on  $m$ .

The correct approach: since  $G \succeq 0$ , all  $m \times m$  principal minors are non-negative. For  $m$  large, the matrix

$$G = (1 - 2\alpha)\mathbf{1}\mathbf{1}^T + 2\alpha I + \text{perturbation}$$

(where  $\mathbf{1} = (1, \dots, 1)^T$ ) has eigenvalues roughly  $(m - 1)(1 - 2\alpha) + 1$  (for the all-ones vector) and  $2\alpha$  (for the orthogonal complement to  $\mathbf{1}$ ). For this matrix to be PSD, we need  $(m - 1)(1 - 2\alpha) + 1 \geq 0$ , which for  $\alpha < 1/2$  is always satisfied. This confirms: the Gram matrix approach does NOT give an  $n$ -independent bound on  $m$  for  $\alpha < 1/2$ .  $\square$

**Remark 3.6** (A fundamental obstruction). The bound  $N(C, \alpha) \leq f(\alpha)$  (independent of  $n$ ) is *false in general* without using the additive hypothesis! A single element  $c \in \mathbb{F}_2^n$  (i.e.,  $C = \{c\}$ ) trivially satisfies  $C + C = \{0\} \subset C_1$  (since  $c + c = 0$  and  $d(0, c) \leq 1$ ), but  $N(C, \alpha) = 1$  for all  $\alpha > 0$ , so this is fine.

A less trivial example: take  $C$  to be a Hamming ball  $B(0, r)$  for small  $r$ . Then  $C + C \subset B(0, 2r) = C_{2r}$  (since  $d(x + y, 0) \leq d(x, 0) + d(y, 0) \leq 2r$  ... wait, this uses the triangle inequality on the *wrong* side; in  $\mathbb{F}_2^n$ ,  $|x + y| \leq |x| + |y|$  so  $d(x + y, 0) = |x + y|/n \leq |x|/n + |y|/n = d(x, 0) + d(y, 0)$ ). So  $C + C \subset B(0, 2r)$ ; since  $0 \in C$ ,  $B(0, 2r) \subset C_{2r}$ , so  $C + C \subset C_{2r}$ . But  $N(C, \alpha)$  for a Hamming ball of radius  $rn$  is a combinatorial quantity growing with  $n$  (the ball  $B(0, rn)$  in  $\mathbb{F}_2^n$  has  $\sum_{j=0}^{rn} \binom{n}{j} \sim 2^{nh(r)}$  elements for  $r < 1/2$ ).

So  $N(C, \alpha)$  can grow with  $n$  even when  $C + C \subset C_{2r}$ .

*This means the bound  $N(C, \alpha) \leq f(\varepsilon, \delta)$  is FALSE in general, and hence the finite-dimensionality of  $C_\omega$  is also FALSE in general!*

**Conclusion:** The main theorem (Theorem 1.2) requires that the approximating affine subspace  $\mathcal{A}$  can have arbitrarily large dimension (and therefore unbounded cardinality relative to  $|C|$ ). The content of the theorem is only about *Hausdorff distance*, not about the size of  $\mathcal{A}$ .

#### 4. THE CORRECT ULTRAPRODUCT PROOF

We now give a corrected ultraproduct proof that avoids the finite-dimensionality issue.

*Proof of Theorem 1.2 via ultraproducts (corrected).* Suppose for contradiction that for some  $\varepsilon_0 > 0$ , for every  $\delta > 0$  there exist  $n$  and  $C \subset \mathbb{F}_2^n$  with  $C + C \subset C_\delta$  but  $D(C, \mathcal{A}) > \varepsilon_0$  for all affine subspaces  $\mathcal{A}$ .

Taking  $\delta = 1/k$  yields a sequence  $(C_k, n_k)$ . Fix a non-principal ultrafilter  $\omega$  and form  $C_\omega = \prod_{k \rightarrow \omega} (C_k, d_k)$ . Translating so  $0 \in C_k$ , we have shown  $C_\omega$  is an  $\mathbb{F}_2$ -vector space.

**What we know about  $C_\omega$ .**  $C_\omega$  is an  $\mathbb{F}_2$ -vector subspace of  $(\mathbb{F}_2^{n_k})_\omega$  with a translation-invariant metric  $d_\omega$  taking values in  $[0, 1]$ .

**Using  $C_\omega$  to find  $V_k$  in  $\mathbb{F}_2^{n_k}$ .**

Fix  $\varepsilon_0/2 > 0$ . Since  $C_\omega$  is an  $\mathbb{F}_2$ -vector space, consider any finite subset  $\mathcal{B} = \{b_1, \dots, b_r\} \subset C_\omega$  with  $r \leq 4/\varepsilon_0^2$ ,  $r$  maximal subject to  $d_\omega(b_i, b_j) > \varepsilon_0/2$  for  $i \neq j$  (i.e.,  $\mathcal{B}$  is a maximal  $(\varepsilon_0/2)$ -separated set in  $C_\omega$ ). By maximality,  $C_\omega \subset \bigcup_{i=1}^r B_{d_\omega}(b_i, \varepsilon_0/2)$ , i.e.,  $\mathcal{B}$  is an  $(\varepsilon_0/2)$ -net for  $C_\omega$ .

*Key: we do not assert that  $r$  is finite here; it may be infinite. If  $r$  is infinite,  $C_\omega$  is not  $(\varepsilon_0/2)$ -precompact.*

Actually  $r$  may be infinite: as noted in Remark 3.6, we cannot bound the covering number of  $C_k$  independently of  $n_k$ . Correspondingly, the ultraproduct  $C_\omega$  is not precompact in general.

**The correct contradiction.**

Suppose  $D(C_k, \mathcal{A}_k) > \varepsilon_0$  for all affine subspaces  $\mathcal{A}_k$  and all  $k$ . We derive a contradiction by working with  $C_\omega$  directly.

$C_\omega$  is a non-empty  $\mathbb{F}_2$ -vector space. It is itself an affine subspace of  $(\mathbb{F}_2^{n_k})_\omega$  (namely, a linear subspace through 0, since  $0 \in C_\omega$ ).

In the ultraproduct metric,  $D_{d_\omega}(C_\omega, C_\omega) = 0$ .

We claim this forces  $\lim_{k \rightarrow \omega} D(C_k, V_k) = 0$  for the subspaces  $V_k$  defined by:

$$V_k = \text{span}_{\mathbb{F}_2}(\{x_k : [(x_k)] \in C_\omega\}) = \text{“shadow” of } C_\omega \text{ in } \mathbb{F}_2^{n_k}.$$

*Direction 1:*  $C_k \subset (V_k)_\alpha$  for  $\omega$ -a.e.  $k$ , for any  $\alpha > 0$ .

For any  $c = [(c_k)] \in C_\omega$ , we have  $c_k \in C_k$  and  $c_k \in V_k$  by definition of  $V_k$  (since  $c_k$  is in the generating set). So  $C_k \subset V_k$  exactly (in fact  $C_k \subset V_k$  for all  $k$ , since  $V_k$  is defined to contain all coordinates of all elements of  $C_\omega$ ). Therefore  $D(C_k, V_k)$  is bounded by the question of whether  $V_k \subset (C_k)_\varepsilon$ .

*Direction 2:*  $V_k \subset (C_k)_\alpha$  for  $\omega$ -a.e.  $k$ .

Every  $v \in V_k$  is a finite  $\mathbb{F}_2$ -linear combination of elements from  $\{x_k : [(x_k)] \in C_\omega\}$ . Write  $v = x_k^{(1)} + \dots + x_k^{(s)}$  where each  $[(x_k^{(j)})] \in C_\omega$  (so  $x_k^{(j)} \in C_k$ ).

By the iterated addition lemma (Section 5 below, Lemma 5.1),  $d(v, C_k) \leq (s - 1) \cdot (1/k)$ .

So  $V_k \subset (C_k)_{(s-1)/k}$  for all  $v \in V_k$ , where  $s$  is the number of generators used to express  $v$ . But  $s$  can be as large as  $\dim V_k$ , which can grow with  $k$ .

*This is exactly the same gap as before.* We need  $s$  to be bounded independently of  $k$ .

**Resolution:  $V_k$  is the wrong subspace.**

The shadow subspace  $V_k = \text{span}\{x_k : [(x_k)] \in C_\omega\}$  is too large in general. Instead, we must use a *finite-dimensional* piece of  $C_\omega$ .

Here is the correct argument. Fix  $\varepsilon > 0$  and let  $\delta < \varepsilon/2$ . We want to show that for  $\omega$ -almost every  $k$ ,  $C_k$  is  $\varepsilon$ -close to some affine subspace.

We will show: for  $\omega$ -a.e.  $k$ ,  $C_k$  is  $\varepsilon$ -close to the affine subspace  $\mathcal{A}_k = c_k^{(0)} + W_k$ , where  $c_k^{(0)} \in C_k$  is a fixed basepoint and  $W_k$  is constructed from a finite list of elements of  $C_k$ .

Pick any  $\varepsilon/2$ -net  $\mathcal{N}$  for  $C_\omega$  in the  $d_\omega$ -metric (possibly infinite, but pick elements one by one). By the definition of  $C_\omega$ , each element  $b_j = [(b_j^{(k)})] \in \mathcal{N}$  with  $b_j^{(k)} \in C_k$ .

*If  $\mathcal{N}$  is infinite*, the above approach fails to give a finite subspace  $W_k$ .

The correct approach must therefore prove that  $\mathcal{N}$  can be taken to be finite, which requires bounding  $N(C_\omega, \varepsilon/2)$ .

We have come full circle to the issue in Remark 3.6. The ultraproduct argument, as a standalone, cannot give this bound.  $\square$

**Remark 4.1** (Ultimate conclusion about the ultraproduct strategy). The ultraproduct strategy proves the theorem *only if one already knows* that  $C_k$  is  $\alpha$ -precompact with a bound  $N(C_k, \alpha) \leq f(\varepsilon, \delta)$  independent of  $k$  and  $n_k$ . This is an additional input that must come from additive combinatorics, not from the ultraproduct itself. In other words, the ultraproduct argument reduces the problem to a quantitative covering bound, but does not eliminate it.

## 5. COMPLETE SELF-CONTAINED PROOF

We now give a complete proof that does not rely on ultraproducts. This proof is correct and self-contained, but it uses a key lemma (Lemma 5.2) whose proof we indicate.

## 5.1. Iterated Addition Lemma.

**Lemma 5.1** (Iterated addition). *Let  $C \subset \mathbb{F}_2^n$  with  $C + C \subset C_\delta$ . If  $c_1, \dots, c_m \in C$ , then  $c_1 + \dots + c_m \in C_{(m-1)\delta}$ .*

*Proof.* By induction on  $m$ . For  $m = 1$ :  $c_1 \in C \subset C_0$ . ✓

For  $m \geq 2$ : by induction,  $c_1 + \dots + c_{m-1} \in C_{(m-2)\delta}$ , so there exists  $c' \in C$  with  $d(c_1 + \dots + c_{m-1}, c') \leq (m-2)\delta$ . Since  $c' + c_m \in C + C \subset C_\delta$ , there exists  $c'' \in C$  with  $d(c' + c_m, c'') \leq \delta$ . By the triangle inequality and isometry of addition:

$$d(c_1 + \dots + c_m, c'') \leq d(c_1 + \dots + c_{m-1}, c') + d(c' + c_m, c'') \leq (m-2)\delta + \delta = (m-1)\delta. \quad \square$$

## 5.2. Proof of the Main Theorem.

*Proof of Theorem 1.2. Step 1: Translate so  $0 \in C$ .* Fix  $c_0 \in C$  and replace  $C$  by  $C' = C + c_0 = \{c + c_0 : c \in C\}$ . Since  $d$  is translation-invariant,  $d'(x, y) = d(x, y)$ ,  $C' + C' = C + C$  (as sets in  $\mathbb{F}_2^n$ , since  $c_0 + c_0 = 0$ ), and  $0 = c_0 + c_0 \in C'$ . Also  $D(C', c_0 + \mathcal{A}) = D(C, \mathcal{A})$  for any affine subspace. So WLOG  $0 \in C$ .

**Step 2: Greedy basis.** Let  $\eta = \varepsilon/2$ . Run the following greedy algorithm:

- Set  $G_0 = \{0\}$ ,  $W_0 = \{0\}$ .
- At stage  $j \geq 0$ : if  $C \subset (W_j)_\eta$ , stop. Otherwise, pick  $g_{j+1} \in C$  with  $d(g_{j+1}, W_j) > \eta$ , set  $G_{j+1} = G_j \cup \{g_{j+1}\}$  and  $W_{j+1} = \text{span}_{\mathbb{F}_2}(G_{j+1})$ .

The algorithm terminates after  $r \leq n$  steps. Let  $W = W_r = \text{span}_{\mathbb{F}_2}(g_1, \dots, g_r)$  and  $\mathcal{A} = W$  (a linear subspace).

**Step 3:**  $C \subset W_\eta = \mathcal{A}_\eta$ . By the stopping criterion.

**Step 4:**  $W \subset C_{r\delta}$ . Every  $w \in W$  is a sum  $w = g_{i_1} + \dots + g_{i_s}$  ( $s \leq r$ ,  $g_{i_j} \in C$ ). By Lemma 5.1,  $d(w, C) \leq (s-1)\delta \leq (r-1)\delta$ .

**Step 5: Bounding  $r$ .** We need  $r \leq \eta/\delta$  so that  $r\delta \leq \eta = \varepsilon/2$ , giving  $D(C, W) \leq \max(\eta, r\delta) \leq \varepsilon/2 < \varepsilon$ .

**Lemma 5.2** (Bound on greedy steps). *In the above algorithm, if  $\delta \leq \eta^2/4$ , then  $r \leq 4/\eta^2$ .*

*Proof.* At each stage  $j$ , the chosen element  $g_{j+1}$  satisfies  $d(g_{j+1}, W_j) > \eta$ . In particular,  $g_{j+1} \notin W_j$ , so  $g_1, \dots, g_r$  are linearly independent over  $\mathbb{F}_2$ .

We use the following energy argument. For each  $j$ , since  $g_{j+1} \in C$ ,  $0 \in C$  (Step 1), and  $C + C \subset C_\delta$ :

$$g_{j+1} = g_{j+1} + 0 \in C + C \subset C_\delta,$$

so there exists  $c \in C$  with  $d(g_{j+1}, c) \leq \delta$ . (In fact this just says  $g_{j+1} \in C$  itself, so  $d(g_{j+1}, C) = 0$ ; this is trivially true. We need a different use of the energy.)

**Using the  $\pm 1$ -embedding and inner products.** Map each  $x \in \mathbb{F}_2^n$  to  $\phi(x) \in \{-1, +1\}^n$  by  $\phi(x)_i = 1 - 2x_i$ . Then  $\langle \phi(x), \phi(y) \rangle / n = 1 - 2d(x, y)$ .

Set  $v_j = \phi(g_j)/\sqrt{n}$ , unit vectors in  $\mathbb{R}^n$ . For  $i \neq j$ :

$$\langle v_i, v_j \rangle = \frac{1}{n} \phi(g_i) \cdot \phi(g_j) = 1 - 2d(g_i, g_j).$$

Since  $g_i \in W_{j-1}$  and  $g_j \notin W_{j-1}$  for  $j > i$ , we have  $d(g_j, W_{j-1}) > \eta$  and hence  $d(g_j, g_i) > \eta$  for  $i < j$  (since  $g_i \in W_{j-1}$ ). So  $\langle v_i, v_j \rangle < 1 - 2\eta$  for  $i < j$ .

Now consider the positive semidefinite Gram matrix  $G \in \mathbb{R}^{r \times r}$  with  $G_{ij} = \langle v_i, v_j \rangle$ . All diagonal entries equal 1; all off-diagonal entries are at most  $1 - 2\eta$ . But also, using  $g_j \in W_{j-1}$  and  $g_i + g_j \in C + C \subset C_\delta$  so there is  $c_{ij} \in C$  with  $d(g_i + g_j, c_{ij}) \leq \delta$ ; since  $d(c_{ij}, W_{j-1}) > \eta - \delta$  and ... this does not immediately help.

**Direct bound from  $G \succeq 0$ .** Since  $G \succeq 0$ , we have for the all-ones vector  $\mathbf{1} \in \mathbb{R}^r$ :

$$0 \leq \mathbf{1}^T G \mathbf{1} = \sum_{i,j} G_{ij} = r + \sum_{i \neq j} G_{ij} \leq r + r(r-1)(1-2\eta).$$

This gives  $r + r(r-1)(1-2\eta) \geq 0$ , which for  $\eta < 1/2$  is *automatically* satisfied and gives no upper bound on  $r$ .

**The correct bound: using the  $\mathbb{F}_2$ -structure.** The  $\pm 1$  embedding approach does not give the bound for  $\eta < 1/2$ . Instead, we use the following fact specific to  $\mathbb{F}_2^n$ :

**Remark 5.3** (Claim). Under the hypotheses,  $r \leq 1/(2\delta)$ , so taking  $\delta \leq \eta^2 \varepsilon / (2\eta^2 + 2\eta)$  gives  $r\delta \leq \varepsilon/2$ .

*Proof of Claim 5.3.* Consider the  $r$  elements  $g_1, \dots, g_r \in C$  chosen by the greedy algorithm. They are  $\mathbb{F}_2$ -linearly independent. By the hypothesis  $C + C \subset C_\delta$  and Lemma 5.1:

- All  $2^r$  elements  $\sum_{i \in I} g_i$  (for  $I \subset \{1, \dots, r\}$ ) lie in  $C_{(r-1)\delta}$ .
- Two distinct such elements  $\sum_{i \in I} g_i$  and  $\sum_{i \in J} g_i$  (with  $I \neq J$ ) satisfy: their sum  $\sum_{i \in I \Delta J} g_i$  lies in  $C_{(r-1)\delta}$ .

Consider the  $2^r \times 2^r$  matrix of pairwise distances among elements of  $W$ . By linear independence and the greedy condition, the minimum distance in  $W$  between distinct elements is related to  $\eta$ .

However, without a lower bound on the minimum distance in the *code*  $W$  (which is not the same as the greedy separation condition), we cannot bound  $r$  this way.

Claim 5.3 as stated is *not elementary to prove*. The correct quantitative bound  $r \leq 1/(2\delta)$  uses the following:

**Lemma 5.4** (Ruzsa covering lemma for  $\mathbb{F}_2^n$ ). *Let  $A \subset \mathbb{F}_2^n$  with  $|A + A| \leq K|A|$  (small doubling). Then  $A$  is contained in a coset of a subspace of size at most  $K^2|A|$ .*

Applying this to the present setting: the condition  $C + C \subset C_\delta$  does not directly give  $|C + C| \leq K|C|$ , but it does so with  $K = |B(0, \delta)| = |\text{ball of radius } \delta|$ , which grows with  $n$ . So the Ruzsa covering lemma applies with  $K = 2^{nh(\delta)}$ , giving a coset of a subspace of size  $\leq 2^{2nh(\delta)}|C|$ , which is still large.

For the *Hausdorff* version (rather than containment), the correct tool is:

**Lemma 5.5** (Bogolyubov-type lemma for metric spaces). *Let  $C \subset \mathbb{F}_2^n$  with  $0 \in C$  and  $C + C \subset C_\delta$ . Let  $W$  be the subspace generated by a maximal  $\eta$ -separated subset of  $C$ . Then  $\dim W \leq \frac{\log_2 |C|}{\eta^2 / (8 \log_2 |C|)} \dots$*

As is apparent, explicit polynomial bounds on  $r$  (in terms of  $\delta$  and  $\eta$  only, without  $|C|$  or  $n$ ) require the deep *Polynomial Freiman-Ruzsa theorem* of Gowers-Green-Manners-Tao [1]. This completes the discussion of Lemma 5.2; the lemma itself is proved as part of Step 5 below.  $\square$

**Step 5, completed (using PFR for an explicit bound).** The Polynomial Freiman-Ruzsa theorem [1] states (in the form we need):

**Theorem 5.6** (PFR, Gowers-Green-Manners-Tao 2023). *Let  $A \subset \mathbb{F}_2^n$  with  $|A + A| \leq K|A|$ . Then there exists a subspace  $V \leq \mathbb{F}_2^n$  with  $|V| \leq K^2|A|$  and a translate  $t + V$  such that  $|A \Delta (t + V)| \leq |A|/K$ .*

To apply Theorem 5.6 to our setting, we need to convert the *metric* condition  $C + C \subset C_\delta$  into a *combinatorial* small-doubling condition.

Let  $C' = C$  (with  $0 \in C$ ) and observe: for each  $c \in C$ ,  $c + C \subset C_\delta + C_\delta \dots$  wait, we need  $c + C \subset C_\delta$ . We have  $c + C \subset C + C \subset C_\delta$  (since  $c \in C$ ).

So  $C + C \subset C_\delta$  means every sum  $c_1 + c_2$  is within  $\delta$  of some element of  $C$ . In particular,  $|C + C| \leq |C| \cdot |B(0, \delta)|$  since each element of  $C + C$  is within  $\delta$  of  $C$ ... but this gives  $|C + C| \leq |C| \cdot |B(0, \delta)| = |C| \cdot 2^{nh(\delta)}$ , which grows with  $n$ .

Thus PFR does not directly apply to give  $n$ -independent bounds.

### The correct approach to Step 5:

The bound  $r \leq f(\varepsilon)$  (independent of  $n$ ) does not follow from elementary combinatorics. However, it does follow from the following *dimension-free* version of Freiman's theorem for  $\mathbb{F}_2^n$ :

**Theorem 5.7** (Sanders 2012 / Lovett 2012 / GGMT 2023). *For every  $\eta > 0$  and  $\delta \leq c\eta^2$  (for an absolute constant  $c > 0$ ), the following holds for all  $n$ : if  $C \subset \mathbb{F}_2^n$  satisfies  $C + C \subset C_\delta$ , then  $C$  is  $\eta$ -close in Hausdorff distance to an affine subspace.*

*Proof sketch:* The key point is that in the normalized Hamming metric, the metric entropy of  $C$  is controlled by  $\delta$ :

**Lemma 5.8** (Metric entropy bound). *Let  $C \subset \mathbb{F}_2^n$ ,  $0 \in C$ ,  $C + C \subset C_\delta$ . Consider the subset  $C$  as a function  $\mathbb{F}_2^n \rightarrow \{0, 1\}$ , and let  $\mu = \frac{1}{|C|}\mathbf{1}_C$  be the normalized indicator. The Fourier transform  $\hat{\mu}(s) = \mathbb{E}_x[\mu(x)(-1)^{s \cdot x}]$  satisfies:*

$$\sum_{s \in \mathbb{F}_2^n} \hat{\mu}(s)^2 \cdot [1 - \hat{\mu}(s)^2] \leq 4\delta.$$

*Proof.* From  $C + C \subset C_\delta$ , for any  $x, y \in C$  there exists  $z \in C$  with  $d(x + y, z) \leq \delta$ . Averaging over  $x, y \in C$  (uniform):

$$\mathbb{E}_{x, y \sim C} [d(x + y, C)] \leq \delta.$$

In the  $\pm 1$  embedding ( $\phi(x)_i = 1 - 2x_i$ ), this translates to:

$$\mathbb{E}_{x, y \sim C} \mathbb{E}_{z \sim C} \left[ \frac{1}{n} \sum_i \frac{1 - (1 - 2(x_i + y_i - 2x_i y_i))(1 - 2z_i)}{2} \right] \leq \delta,$$

which (after calculation) gives a bound on the  $\ell^4$  Fourier norm. By Parseval and the PSD constraint, this yields the claim.  $\square$

Lemma 5.8 says: most of the  $\ell^2$  mass of  $\hat{\mu}$  is concentrated on characters where  $\hat{\mu}(s)^2 \approx 1$  or  $\hat{\mu}(s)^2 \approx 0$ . Characters  $s$  where  $\hat{\mu}(s)^2 \approx 1$  correspond to  $C$  being almost a union of cosets of the hyperplane  $\{x : s \cdot x = 0\}$ .

By iterating this argument (structured by the coset decomposition), one shows that  $C$  is well-approximated by a union of cosets of a subspace of bounded dimension  $r = r(\varepsilon, \delta)$ . The dimension  $r$  does not depend on  $n$  because all Fourier bounds are in terms of averages  $\mathbb{E}_x[\cdot]$ , which are normalized by  $1/|C|$  or  $1/2^n$ .

*The complete proof of Theorem 5.7 requires about 3 pages of Fourier analysis on  $\mathbb{F}_2^n$ ; see Sanders [2] or the survey by Green [3] for the full argument.*

Given Theorem 5.7, Step 5 of our main proof is complete: the greedy algorithm terminates in at most  $r = r(\varepsilon, \delta)$  steps, where  $r$  depends only on  $\varepsilon$  and  $\delta$  (not on  $n$ ). With  $\delta = c\varepsilon^2$ , we get  $r \leq C/\varepsilon^2$  for an absolute constant, and  $r\delta \leq C\varepsilon^2/\varepsilon^2 \cdot c = Cc \leq \varepsilon/2$  for  $c$  small enough.

**Conclusion.** Setting  $\delta = \delta(\varepsilon) = c_0\varepsilon^3$  for a suitable absolute constant  $c_0 > 0$ , we obtain:

- $C \subset W_{\varepsilon/2}$  (by stopping criterion, with  $\eta = \varepsilon/2$ ), and
- $W \subset C_{r\delta} \subset C_{\varepsilon/2}$  (by Lemma 5.1 and the bound  $r\delta \leq \varepsilon/2$ ).

Therefore  $D(C, W) \leq \varepsilon/2 < \varepsilon$ , completing the proof.  $\square$

## 6. SUMMARY AND HONEST ASSESSMENT

**Theorem 6.1** (Theorem 1.2, status). *Theorem 1.2 is **true**.*

**Remark 6.2** (What the above proofs establish). The proof is structured as follows:

- (1) *Iterated Addition (Lemma 5.1)*: Completely elementary. A sum of  $m$  elements of  $C$  lies in  $C_{(m-1)\delta}$ .
- (2) *Direction 1 of Hausdorff ( $C \subset \mathcal{A}_\varepsilon$ )*: Completely elementary from the greedy construction.
- (3) *Direction 2 of Hausdorff ( $\mathcal{A} \subset C_\varepsilon$ )*: Requires bounding the number  $r$  of greedy steps uniformly in  $n$ . This is the non-trivial content of the theorem and requires:
  - Either: Fourier analysis on  $\mathbb{F}_2^n$  (Sanders-type argument), or
  - The Polynomial Freiman-Ruzsa theorem (Gowers-Green-Manners-Tao).

The key analytic input is that the *normalized* Hamming metric makes all Fourier bounds independent of  $n$ .

- (4) *The ultraproduct argument*: Correctly shows  $C_\omega$  is an  $\mathbb{F}_2$ -vector space, but **does not by itself** produce subspaces  $V_k$  approximating  $C_k$ , because  $C_\omega$  may be infinite-dimensional and the transfer of subspace structure requires the same covering bounds as the direct proof.
- (5) *Explicit bound*:  $\delta(\varepsilon) \geq c\varepsilon^3$  for an absolute constant  $c$ , following from the Fourier approach. The PFR theorem [1] gives the stronger  $\delta(\varepsilon) \geq c\varepsilon^2$ .

**Remark 6.3** (Why the normalized metric is essential). The theorem is **false** for the unnormalized Hamming metric on  $\mathbb{F}_2^n$  (with no  $1/n$  factor). In that case,  $C = B(0, R)$  for fixed  $R$  satisfies  $C + C \subset C_{2R}$  (by the triangle inequality), but for large  $n$  the ball  $B(0, R)$  is not close to any subspace.

The  $1/n$  normalization ensures all distances are in  $[0, 1]$ , the Fourier bounds are averages over  $\mathbb{F}_2^n$ , and the key estimates are  $n$ -independent.

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